

Natural Boundaries Survive Nonlinear Feedback

Moving-parameter partial theta series,
sharp half-plane realization,
and the failure of angular Borel summation

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Abstract

For an integer $d \geq 2$, consider the formally well-defined equation

$$\hat{U}_d(q) = \sum_{j \geq 1} q^j \exp(j^d \hat{U}_d(q))$$

and its compositional inverse $\hat{Q}_d$. We prove that the canonical actual inverse Q_d , holomorphic in a sufficiently small left half-disc and smooth up to its diameter, has that entire imaginary diameter as a natural boundary. The obstruction survives analytic changes of the moving parameter, analytic changes of the level set, and finite modifications of the feedback kernel. The main tool is an explicit large-order theorem: periodic geometric moments retain factorial growth after substitution along an arbitrary holomorphic curve. A mean-square argument treats tied nearest poles without assuming a unique dominant singularity.

For $d = 2$, this resolves the angular-continuation question left open in the repository's negative-ray summation article: both inverse and forward formal series are finely Borel summable on the negative ray, but neither is Borel summable in any open angular neighborhood of that direction. In particular, an entire Borel transform does not supply an angular summation sector. We also give convergent exponential blocks, uniform closed-half-disc remainders for every $d \geq 2$, an explicit optimized error scale, finite verification programs, and further research questions.

Proof status. The new assertions are accompanied by conventional mathematical proofs. They have not been checked in Lean or independently peer reviewed. The contribution is a model-specific natural-boundary and non-summability theorem, not a claimed solution of a named longstanding conjecture. The literature and repository comparison is targeted; global priority is not asserted.

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1 The question resolved and the contribution boundary

1.1 The precise repository problem

The ProveIt transseries documents distinguish formal coefficient calculations, actual analytic realizations, remainder estimates, and summability. Those levels must not be conflated. In the feedback family

$$\widehat{U}(q) = \sum_{j \geq 1} q^j e^{\lambda_j \widehat{U}(q)}, \quad \lambda_j \geq 0, \quad (1.1)$$

several recent repository articles establish regularity classifications and negative-direction realizations [2, 3, 4]. The most direct predecessor, *Negative-Ray Summation of Countable Exponential-Feedback Transseries*, proves fine negative-ray summability at the threshold $\lambda_j = O((j \log(j+1))^2)$. It explicitly does not claim angular summability or a maximal continuation domain [4].

This paper addresses the missing angular question for the quadratic model $\lambda_j = j^2$, and proves a natural-boundary theorem for the larger family $\lambda_j = j^d$, $d \in \mathbb{N}$, $d \geq 2$. The answer in the quadratic case is *negative*: fine summation cannot be enlarged to an open angular interval containing the negative direction. The reason is a natural boundary of the actual inverse, not failure of the Borel germ to continue to finite points.

Editorial note (ProveIt, 2026-09-29). For $d = 2$ this question had already been answered negatively, independently, by an earlier-filed package that this article could not see [5]; see the editorial note after Theorem 1.2. The cases $d \geq 3$ are new to the repository.

The source snapshot used for the final comparison is

ebd8344bca77d8352cd745b7df374618a290a029.

The principal files are under `Analysis/Transseries/docs/series-and-transseries/`. The source inventory and exact blob identifiers are recorded in the package's provenance file. No repository files were changed. The canonical large volume was not audited line by line, and no assertion about the entire repository's formal proof status is needed here.

1.2 Main statements

Write

$$\Theta_d(q, u) = \sum_{j \geq 1} q^j e^{j^d u}, \quad H_r^- = \{u : |u| < r, \operatorname{Re} u < 0\}. \quad (1.2)$$

For $|q| < 1$, the series and all its mixed derivatives extend continuously to $\operatorname{Re} u = 0$. The defining identity for the actual inverse is

$$\Theta_d(Q_d(u), u) = u, \quad Q_d(u) = u + O(u^2). \quad (1.3)$$

Theorem 1.1 (Natural boundary of the inverse and forward realization). *For each integer $d \geq 2$, there is $r > 0$ such that:*

- (i) *Equation (1.3) has a unique small solution Q_d on H_r^- . It is holomorphic there and C^∞ up to the imaginary diameter, including the origin.*
- (ii) *No point of $\{it : |t| < r\}$ admits a holomorphic continuation of Q_d through the diameter. Thus this whole boundary arc is a natural boundary.*
- (iii) *After reducing r , Q_d is one-to-one, and its inverse U_d is holomorphic on $\Omega_d = Q_d(H_r^-)$. The smooth arc $\Gamma_d(t) = Q_d(it)$ is a natural boundary of U_d .*

(iv) Uniformly as real $t \rightarrow 0$,

$$\Gamma_d(t) = it + 2t^2 - i\left(\frac{9}{2} - 2^d\right)t^3 + O(t^4). \quad (1.4)$$

Neither canonical realization can be continued to a sector about the negative axis with opening strictly greater than π .

The phrase “natural boundary” here refers to the displayed boundary arc. It is not a claim that the circular edge $|u| = r$ is singular or that the small half-disc is a globally maximal domain.

Theorem 1.2 (Fine but not angular in the quadratic case). *For $d = 2$, the formal series $\hat{Q}_2$ and $\hat{U}_2$ have entire integrated Borel transforms and are finely Borel summable along the negative ray. Their sums are Q_2 and U_2 near that ray. Neither formal series is 1-summable in an open angular interval containing direction π .*

The same fine-versus-angular conclusion holds when finitely many of the nonnegative slopes j^2 in (1.1) are changed to other finite nonnegative values.

Precise summability conventions are given in Section 10. Fine summability means a fixed-width Borel tube; angular summability means a Borel sector with locally uniform exponential bounds. These are different requirements.

Editorial note (ProveIt, 2026-09-29). The case $d = 2$ of Theorems 1.1 and 1.2 was proved independently in `Natural_Boundaries_Quadratic_Exponential_Feedback/` [5] (its Theorem `thm:main`), filed earlier and not visible from this article’s snapshot, with the explicit radius $1/32$, the bound $|Q' - 1| < 1/3$ and the further conclusion that the image arc is nowhere real-analytic. Its Theorem `thm:rigidity` is the case $d = 2$ of Theorem 4.1, and its Theorem `thm:rational-main` extends the $d = 2$ conclusions to rational amplitude series. The exact coefficients at $d = 2$ agree with that package’s recorded ones (through degree 16). The cases $d \geq 3$ answer that package’s question “Higher-degree and nonpolynomial slopes” for integer d ; noninteger powers remain open.

1.3 What is new, inherited, and deliberately not claimed

The principal new mechanism is the moving-curve moment theorem in Section 4. A fixed-parameter lacunary-series theorem alone does not settle (1.3): its parameter q changes with u , so one must exclude cancellation by that motion. The resulting natural boundary and angular obstruction are the proposed research contribution.

Editorial note (ProveIt, 2026-09-29). At $d = 2$ the moving-curve theorem, the natural boundary and the angular obstruction are also in [5] (see the editorial note after Theorem 1.2); they are new to the repository for $d \geq 3$.

The coefficient/Gevrey classification and the positive fine-summability result are earlier repository results. Sections 8–10 recover the quadratic positive result in a specialized form to make the contrast transparent. General Laplace summation and its nonlinear closure are classical [8, 9]. Partial-theta boundary asymptotics, rational boundary points, and discrete Fourier transforms also have a substantial literature [10, 11]. The new argument here concerns a geometric parameter constrained by a nonlinear level-set equation, not a new discovery of theta-series boundary phenomena.

We do not infer failure of resurgence merely from a natural boundary of the summed function. Nor do we identify a Stokes constant from an upper remainder bound. For $d > 2$, the natural-boundary assertion is about the canonical literal realization; no unsupported identification with every possible higher-order summation procedure is made.

2 Formal series and actual half-plane solutions

2.1 Formal existence is not analytic existence

Let $\mathbb{C}[[q]]$ have its coefficientwise, or q -adic, topology. If V, W have zero constant term and agree modulo q^n , then

$$\sum_{j \geq 1} q^j e^{j^d V} \equiv \sum_{j \geq 1} q^j e^{j^d W} \pmod{q^{n+1}}.$$

Each coefficient uses only finitely many j . Iteration beginning with zero therefore yields a unique formal solution $\hat{U}_d = q + O(q^2)$. It has a unique compositional inverse $\hat{Q}_d = u + O(u^2)$, and substitution in the formal identity gives

$$\sum_{j \geq 1} \hat{Q}_d(u)^j e^{j^d u} = u.$$

No analytic implicit-function theorem at $(q, u) = (0, 0)$ is being applied: the kernel need not be holomorphic on a full product neighborhood of that point. This distinction also matters when applying general Gevrey implicit-function theorems such as [12].

Direct coefficient comparison gives

$$\hat{U}_d(q) = q + 2q^2 + (2^d + \frac{7}{2})q^3 + O(q^4), \quad (2.1)$$

$$\hat{Q}_d(u) = u - 2u^2 + (\frac{9}{2} - 2^d)u^3 + O(u^4). \quad (2.2)$$

For $d = 2$ the next coefficients are

$$\hat{U}_2(q) = q + 2q^2 + \frac{15}{2}q^3 + \frac{107}{3}q^4 + \frac{1555}{8}q^5 + O(q^6), \quad (2.3)$$

$$\hat{Q}_2(u) = u - 2u^2 + \frac{1}{2}u^3 - \frac{2}{3}u^4 - \frac{29}{8}u^5 + O(u^6). \quad (2.4)$$

These finite identities are checked independently in the accompanying program.

2.2 A uniform small root, including boundary values

Lemma 2.1 (Smooth half-plane implicit root). *Let f be holomorphic near zero, with $f(0) = 0$. For every fixed $d \geq 2$, a sufficiently small closed left half-disc admits a unique small root $q = Q_f(u)$ of*

$$\Theta_d(q, u) = f(u). \quad (2.5)$$

It is holomorphic in the open half-disc, C^∞ up to its diameter, and $Q_f(u) = f(u) + O(|uf(u)| + |f(u)|^2)$. The root vanishes exactly where f does. For $f(u) = u$, its boundary Taylor coefficients are those of $\hat{Q}_d$.

Proof. Separate the first term and use the fixed point

$$q = T_u(q) := e^{-u} \left(f(u) - \sum_{j \geq 2} q^j e^{j^d u} \right).$$

On $|q| \leq R = 1/8$, $\operatorname{Re} u \leq 0$, the tail and its q -derivative are bounded by

$$\frac{R^2}{1-R}, \quad \frac{1}{(1-R)^2} - 1.$$

Choose $|u| \leq 1/16$ and then shrink the radius so that $|f(u)| \leq R/4$. The displayed bounds show that T_u maps the closed R -disc strictly into itself and has Lipschitz constant less than $1/2$.

Uniform iteration gives the root and holomorphic dependence for $\operatorname{Re} u < 0$. Comparison with $T_u(0)$ gives $|Q_f(u)| \leq 2e^{|u|}|f(u)|$; reinsertion gives the stated expansion.

For each fixed pair of derivative orders, the mixed derivatives of the kernel on a smaller q -disc are dominated by a convergent series of the form $C \sum_{j \geq 1} j^M R_1^j$, $R_1 < 1$. Thus every mixed derivative is continuous up to $\operatorname{Re} u = 0$. The denominator $1 - \partial_q T_u$ stays bounded away from zero. Repeated implicit differentiation therefore extends all derivatives of the root continuously to the diameter. This also proves the stated C^∞ regularity without assuming holomorphy across the diameter.

Since $\Theta_d(0, u) = 0$, uniqueness implies the assertion about zeros. Differentiation at zero, or finite Taylor comparison, identifies each finite jet with the unique formal solution of the inverse equation. $\square$

The C^∞ conclusion does not assert convergence of the infinite Taylor series. Smoothness at a boundary and holomorphic continuation through that boundary are very different properties.

3 Periodic moments and their pole geometry

3.1 A finite Fourier transform produces all relevant poles

Let (c_j) be a nonzero complex sequence of period b . Write

$$\gamma_\ell = \frac{1}{b} \sum_{s=0}^{b-1} c_s e^{-2\pi i \ell s/b}, \quad c_j = \sum_{\ell=0}^{b-1} \gamma_\ell e^{2\pi i \ell j/b}. \quad (3.1)$$

For $0 < |q_0| < 1$, define the geometric moments

$$M_m = \sum_{j \geq 1} c_j q_0^j j^m \quad (m \geq 0). \quad (3.2)$$

Choose one value of $\log q_0$; changing it only relabels the following poles.

Lemma 3.1 (Moment singularities). *The exponential generating function of (3.2) is meromorphic:*

$$\mathcal{M}(z) = \sum_{m \geq 0} \frac{M_m}{m!} z^m = \sum_{\ell=0}^{b-1} \gamma_\ell \frac{q_0 e^{z+2\pi i \ell/b}}{1 - q_0 e^{z+2\pi i \ell/b}}. \quad (3.3)$$

Its nonremovable poles are precisely

$$\mathcal{A} = \{-\log q_0 - 2\pi i \ell/b + 2\pi i k : \gamma_\ell \neq 0, k \in \mathbb{Z}\}. \quad (3.4)$$

At $a \in \mathcal{A}$, the principal part is $\gamma_a/(a - z)$, where γ_a is the corresponding Fourier coefficient. Let

$$\rho = \min_{a \in \mathcal{A}} |a|, \quad J = \{a \in \mathcal{A} : |a| = \rho\}.$$

Then J has one or two elements. For some $R > \rho$,

$$M_m = m! \sum_{a \in J} \frac{\gamma_a}{a^{m+1}} + O(m! R^{-m}). \quad (3.5)$$

In particular $|M_m| \leq K m! \rho^{-m}$ for all m , with a fixed K .

Proof. For z close to zero, absolute convergence permits summation of each geometric series after (3.1); this gives (3.3). Distinct nonzero Fourier modes have disjoint pole progressions, so their poles cannot cancel. All poles have real part $\alpha = \log(1/|q_0|) > 0$. Minimizing their modulus

therefore minimizes the absolute value of their imaginary part; at most one point of each sign can attain that minimum. Moreover a nearest pole has imaginary part of modulus at most π , because each individual progression has step $2\pi i$.

Remove the principal parts of the poles in J and choose R strictly between ρ and the next pole modulus. Cauchy's coefficient estimate on a circle of radius R gives (3.5), with an arbitrarily slight reduction of R if needed. The final uniform bound follows, including finitely many initial indices by enlarging K . $\square$

3.2 Why rational boundary points are enough

At a rational imaginary point

$$u_0 = 2\pi ia/b, \quad a \in \mathbb{Z}, \quad b \in \mathbb{N}_{>0}, \quad (3.6)$$

the sequence $c_j = e^{j^d u_0}$ is periodic in j and never zero. Indeed, $(j+b)^d - j^d$ is divisible by b . No estimate for a Gauss sum is required for the existence of a pole: a nonzero periodic sequence has at least one nonzero Fourier coefficient.

For the quadratic case with b odd and $\gcd(a, b) = 1$, one additionally has

$$|\gamma_\ell| = b^{-1/2} \quad (0 \leq \ell < b). \quad (3.7)$$

To check this, square the modulus of the unnormalized Fourier sum, put $s = t + h$, and sum first over t . The inner sum is zero unless $2ah = 0 \pmod{b}$, which forces $h = 0$. Its remaining value is b . Thus all b pole progressions occur in this case. We use this identity only for explicit diagnostics, not to bypass possible vanishing Fourier modes in the general theorem.

4 The moving-curve large-order theorem

The central problem is whether an analytic choice of q depending on u can cancel all the boundary singularities. The next result rules this out. It is a statement about prescribed holomorphic curves, not an unproved asymptotic formula for the derivatives of the implicit solution itself.

Let h be holomorphic near zero with $h(0) = 0$, and let $h_1 = h'(0)$. Consider

$$F(v) = \sum_{j \geq 1} c_j q_0^j \exp(j^d v + j h(v)), \quad \operatorname{Re} v \leq 0, \quad (4.1)$$

for sufficiently small v . This is the substitution $q(v) = q_0 e^{h(v)}$. All derivatives extend continuously to zero from the left half-plane. Write $f_n = F^{(n)}(0)/n!$ for these boundary coefficients.

Theorem 4.1 (Persistence of the nearest-pole contribution). *With the notation of Lemma 3.1, for $d = 2$,*

$$f_n = \frac{(2n)!}{n!} \left(\sum_{a \in J} \frac{\gamma_a e^{a h_1/2}}{a^{2n+1}} + o(\rho^{-2n}) \right). \quad (4.2)$$

For each integer $d \geq 3$,

$$f_n = \frac{(dn)!}{n!} \left(\sum_{a \in J} \frac{\gamma_a}{a^{dn+1}} + o(\rho^{-dn}) \right). \quad (4.3)$$

Thus a holomorphic motion changes the leading quadratic amplitude by a nonzero factor. For $d \geq 3$, it does not even change the leading amplitude.

4.1 An exact coefficient identity

Expand only a fixed derivative order. Absolute convergence of each moment justifies the finite coefficient calculation

$$f_n = \sum_{m=0}^n \sum_{k=0}^m \frac{[v^m]h(v)^k}{k!(n-m)!} M_{d(n-m)+k}. \quad (4.4)$$

For $k = 0$, only $m = 0$ contributes. The term $(m, k) = (0, 0)$ is $M_{dn}/n!$. The identity does not assert convergence of a Taylor expansion of F .

Choose $H, R_h > 0$ such that $|[v^\ell]h(v)| \leq H R_h^{-\ell}$ for all $\ell \geq 1$. Then for $m \geq k \geq 1$,

$$|[v^m]h(v)^k| \leq H^k R_h^{-m} \binom{m-1}{k-1}. \quad (4.5)$$

This follows by comparing h with $H(v/R_h)/(1-v/R_h)$ coefficientwise in absolute value.

4.2 The factorial estimate that controls all indices

For integers $0 \leq s \leq N$,

$$\frac{N!}{(N-s)!} \geq (N/e)^s. \quad (4.6)$$

The geometric mean of the largest s integers among $1, \dots, N$ is at least the geometric mean of all N , and $N! \geq (N/e)^N$. Consequently

$$\frac{n!}{(n-m)!} \frac{(dn-dm+k)!}{(dn)!} \leq n^m \left(\frac{e}{dn} \right)^{dm-k}. \quad (4.7)$$

This bound is valid even when m is close to n . An estimate proved only for fixed m would not justify the theorem.

Normalize the absolute value of a summand in (4.4) by

$$L_n = \frac{(dn)!}{n!} \rho^{-dn}.$$

Using (4.5), (4.7), and the uniform moment bound, and putting $m = k + r$, gives a bound of the form

$$K \frac{X_n^k}{k!} \binom{k+r-1}{r} Y_n^r, \quad X_n = C_1 n^{-(d-2)}, \quad Y_n = C_2 n^{-(d-1)}. \quad (4.8)$$

Here C_1, C_2 are fixed positive constants. For example one may take $C_2 = (e\rho/d)^d/R_h$ and $C_1 = C_2 H d/(e\rho)$. For $Y_n < 1$,

$$\sum_{k \geq 1} \sum_{r \geq 0} \frac{X_n^k}{k!} \binom{k+r-1}{r} Y_n^r = \exp\left(\frac{X_n}{1-Y_n}\right) - 1. \quad (4.9)$$

All extensions of finite sums to infinite sums here are made only for nonnegative upper bounds.

4.3 Completion of the asymptotic proof

Proof of Theorem 4.1. When $d \geq 3$, $X_n \rightarrow 0$ and $Y_n \rightarrow 0$. Equations (4.8)–(4.9) show that every term other than $(0, 0)$ contributes $o(L_n)$ in total. Apply (3.5) to $M_{dn}/n!$ to obtain (4.3).

For $d = 2$, X_n is constant and $Y_n = O(n^{-1})$. The part with $r \geq 1$ is bounded, after normalization, by

$$K \left\{ \exp\left(\frac{X_n}{1-Y_n}\right) - \exp(X_n) \right\} = O(n^{-1}).$$

It remains to evaluate the diagonal $m = k$, on which $[v^k]h(v)^k = h_1^k$. A principal part $\gamma_a/(a - z)$ contributes

$$\frac{(2n)!}{n!} \frac{\gamma_a}{a^{2n+1}} \sum_{k=0}^n \frac{(ah_1)^k}{k!} \frac{n!}{(n-k)!} \frac{(2n-k)!}{(2n)!}.$$

For each fixed k , the product of factorial ratios tends to 2^{-k} . The global bound supplies a summable majorant of the form $C^k/k!$. Dominated convergence therefore makes the sum tend to $e^{ah_1/2}$.

For the analytic remainder in (3.5), each fixed diagonal index gives $o(L_n)$, because $R > \rho$. The same summable majorant permits dominated convergence over the diagonal. The off-diagonal part was already $O(L_n/n)$. Summing over the finite set J proves (4.2). $\square$

Remark 4.2 (Scope of the amplitude formula). The factor $e^{ah_1/2}$ is nonzero regardless of the complex value of h_1 . This does not mean that two nearest-pole contributions can never cancel at an individual order. The next section proves that they cannot cancel at every large order. Higher Taylor coefficients of the prescribed curve enter lower orders of the large-order expansion; no formula for them is needed here.

5 Tied poles and the local nonextension theorem

Lemma 5.1 (A finite oscillatory sum cannot disappear). *If $z_1, \dots, z_s$ are distinct complex numbers of modulus one and not all A_j vanish, then*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \left| \sum_{j=1}^s A_j z_j^n \right|^2 = \sum_{j=1}^s |A_j|^2 > 0. \quad (5.1)$$

In particular the inner sum has modulus bounded below by a positive constant along an infinite subsequence.

Proof. Expand the square. Every off-diagonal average is a finite geometric average with ratio $z_j \bar{z}_k \neq 1$, so tends to zero. The diagonal terms are constant. The subsequence assertion follows immediately. $\square$

Proposition 5.2 (No analytic cancellation along a moving curve). *The function (4.1) has no holomorphic continuation through $v = 0$ in either of the following cases:*

- (i) $d = 2$ and $0 < |q_0| < 1$;
- (ii) $d \geq 3$ and $0 < |q_0| < e^{-d}$.

The same conclusion holds after adding any function holomorphic near zero.

Proof. The set J has at most two elements. For $d = 2$, distinct $a_1, a_2 \in J$ cannot satisfy $a_1^2 = a_2^2$, because that would give $a_1 = -a_2$, whereas both have the same positive real part. For $d \geq 3$, the assumed bound gives $\operatorname{Re} a > d$. Every nearest pole has $|\operatorname{Im} a| \leq \pi$, so $|\arg a| < \pi/d$. Two distinct nearest poles therefore have distinct d th powers: their argument difference is strictly between $-2\pi/d$ and $2\pi/d$.

In Theorem 4.1, divide the expression in parentheses by ρ^{-dn} . Its principal part is a finite sum

$$\sum_{a \in J} A_a z_a^n, \quad z_a = (\rho/a)^d,$$

where $A_a = \gamma_a e^{ah_1/2}/a$ for $d = 2$ and $A_a = \gamma_a/a$ for $d \geq 3$. Every A_a is nonzero and the z_a are distinct. Lemma 5.1 and the $o(1)$ error imply, for infinitely many n ,

$$|f_n| \geq c \frac{(dn)!}{n! \rho^{dn}} \quad (5.2)$$

with a fixed $c > 0$. Stirling's formula shows that the n th root of the right-hand side is asymptotic, up to the factor $c^{1/n}$, to

$$d^d e^{1-d} \rho^{-d} n^{d-1},$$

which diverges. A holomorphic germ has geometrically bounded Taylor coefficients. Its Taylor coefficients would have to equal the boundary coefficients f_n , a contradiction. Addition of a holomorphic function changes the coefficients by at most a geometric sequence and cannot remove (5.2). $\square$

Corollary 5.3 (Rational-point obstruction for partial theta level sets). *Let $u_0 = 2\pi ia/b$, and let $q(u)$ be holomorphic near u_0 , with $q(u_0) \neq 0$. Assume $|q(u_0)| < 1$ for $d = 2$, or $|q(u_0)| < e^{-d}$ for $d \geq 3$. Then the left-hand function*

$$u \longmapsto \Theta_d(q(u), u)$$

cannot extend holomorphically through u_0 .

Proof. Put $v = u - u_0$, $q_0 = q(u_0)$, and choose the local logarithm $h(v) = \log(q(u_0 + v)/q_0)$. The periodic coefficients are $c_j = e^{j^d u_0}$. Apply Proposition 5.2. $\square$

The small bound e^{-d} in the higher-degree case is a convenient sufficient condition that avoids resonances between nearest-pole d th powers. It is not claimed to be an optimal bound. No such extra smallness is needed for $d = 2$.

6 Natural boundaries of nonlinear implicit solutions

Theorem 6.1 (Analytic level sets inherit a natural boundary). *Let f be holomorphic near zero with $f(0) = 0$ and $f \not\equiv 0$. For each integer $d \geq 2$, the small solution Q_f of (2.5) has an entire imaginary diameter as a natural boundary, after the radius is made sufficiently small.*

Proof. Use Lemma 2.1, shrinking the radius so that $|Q_f| < 1$ for $d = 2$ and $|Q_f| < e^{-d}$ for $d \geq 3$. At every rational imaginary boundary point u_0 for which $f(u_0) \neq 0$, the root $Q_f(u_0)$ is nonzero. If Q_f extended holomorphically through this point, Corollary 5.3 would say that $\Theta_d(Q_f(u), u)$ cannot do so. But that function equals the holomorphic function $f(u)$ on the left. This is a contradiction.

The exceptional zeros of f are isolated. Rational imaginary points not among those zeros are dense in the diameter. If continuation existed through any point of the diameter, its open neighborhood would contain one of these already forbidden rational points. Restricting the same continuation to a smaller neighborhood of that rational point gives the final contradiction. $\square$

This proves Theorem 1.1(i)–(ii). In particular, the boundary is not merely a sequence of inaccessible rational points: density and openness of holomorphic continuation give nonextension at every point of the arc.

6.1 Finite-core robustness and periodic amplitudes

Theorem 6.2 (Robustness under analytic finite-core changes). *Let (b_j) be a nonzero periodic sequence, $J_0 \geq 1$, and suppose*

$$K(q, u) = A(q, u) + \sum_{j \geq J_0} b_j q^j e^{j^d u}, \quad (6.1)$$

where A is holomorphic near $(0, 0)$, $K(0, u) = 0$, and $\partial_q K(0, 0) \neq 0$. If f is as in Theorem 6.1, the small solution of $K(q, u) = f(u)$ is holomorphic in a left half-disc, smooth to its diameter, and has that diameter as a natural boundary.

Proof. The uniform geometric derivative bounds used in Lemma 2.1 still hold, with a constant accounting for $\max |b_j|$ and the analytic term. The nonzero linear derivative supplies the small implicit root, either by a normalized contraction or by its usual uniform proof.

At a rational imaginary point, the product $b_j e^{j^d u_0}$ is a nonzero periodic sequence. Were the root analytic there, the omitted finitely many terms and the composition $A(q(u), u)$ would be holomorphic. Adding back those finitely many terms would contradict Proposition 5.2. The same density argument completes the proof. $\square$

Corollary 6.3 (Changing finitely many slopes does not remove the boundary). *If the slopes in (1.1) equal j^d for all sufficiently large j , and all the remaining slopes are finite, the canonical small inverse has the same natural-boundary conclusion. Nonnegativity is not needed for this boundary assertion, although it is convenient for the summability statements.*

Proof. The difference between the altered kernel and Θ_d is a finite sum of entire functions of (q, u) . The altered kernel still has linear q -derivative equal to one at $(0, 0)$. Apply Theorem 6.2. $\square$

7 The forward boundary and its geometry

Proof of Theorem 1.1(iii)–(iv). The derivative $Q'_d(u)$ extends continuously to zero and equals one there. On a sufficiently small closed half-disc, $|Q'_d - 1| \leq 1/2$. For two points u, v in that convex set, integration along their segment gives

$$|Q_d(v) - Q_d(u) - (v - u)| \leq \frac{1}{2}|v - u|.$$

Thus Q_d is injective, with a continuous inverse on its image, and is biholomorphic in the interior. The arc $\Gamma_d(t) = Q_d(it)$ has nonzero velocity $iQ'_d(it)$.

Suppose $U_d = Q_d^{-1}$ extended holomorphically through $q_0 = \Gamma_d(t_0)$. Its interior derivative tends to $1/Q'_d(it_0) \neq 0$, so the extension has nonzero derivative at q_0 . The holomorphic inverse function theorem would extend Q_d through it_0 , contradicting Theorem 6.1. Hence the whole arc is a natural boundary for the forward realization.

The finite Taylor expansion (2.2) is valid along the diameter by the proved smoothness. Substitution $u = it$ gives (1.4). For $d = 2$,

$$\Gamma_2(t) = it + 2t^2 - \frac{i}{2}t^3 - \frac{2}{3}t^4 - \frac{29i}{8}t^5 + O(t^6). \quad (7.1)$$

The inverse's singular diameter approaches directions $\pm\pi/2$ exactly; the forward singular arc approaches them with a positive real displacement $2t^2 + O(t^4)$. Every sector about the negative axis of opening greater than π contains sufficiently small nonzero points of these boundary arcs. An extension to such a sector would contradict their nonextension. $\square$

Remark 7.1 (A geometric rather than algebraic obstruction). All finite-order formal inversions remain valid. The obstruction appears when infinitely many exponential feedback terms are realized analytically. A smooth boundary graph may therefore be visible to arbitrarily long finite jets while remaining impossible to cross holomorphically.

8 Convergent exponential blocks and uniform remainders

Natural boundaries do not prevent accurate evaluation on the permitted side. We give a constructive realization which also connects the theorem back to transseries. Set $P_d(u) = e^u Q_d(u)$. Then

$$P_d = u - \sum_{j \geq 2} P_d^j e^{(j^d - j)u}. \quad (8.1)$$

Introduce a temporary variable t in place of the first u on the right. The solution has a block expansion

$$P_d(u) = \sum_{k \geq 1} u^k H_{d,k}(u), \quad H_{d,k}(u) = \sum_{\beta} h_{d,k,\beta} e^{\beta u}, \quad (8.2)$$

where each inner sum is finite and each coefficient is an integer.

Proposition 8.1 (Tree support and a convergent block certificate). *Let*

$$C(t) = \frac{1 + t - \sqrt{1 - 6t + t^2}}{4} = \sum_{k \geq 1} c_k t^k, \quad \rho_* = 3 - 2\sqrt{2}. \quad (8.3)$$

Then

$$0 \leq \beta \leq k^d - k, \quad \sum_{\beta} |h_{d,k,\beta}| \leq c_k \leq \rho_*^{-k}. \quad (8.4)$$

The block expansion converges normally on every closed left half-disc $|u| \leq r_0 < \rho_$. Its tail after $k = K$ has modulus at most*

$$\frac{(|u|/\rho_*)^{K+1}}{1 - |u|/\rho_*}. \quad (8.5)$$

Proof. Expand (8.1) recursively as plane rooted trees whose internal vertices have arity $j \geq 2$. A tree with k leaves contributes a sign, and the frequency is the sum of $j^d - j$ over internal vertices. If their arities are j_v , then $\sum_v (j_v - 1) = k - 1$. The polynomial $(s + 1)^d - (s + 1)$ has nonnegative coefficients and vanishes at $s = 0$; its value at a sum of nonnegative arguments is at least the sum of its values. Hence

$$\sum_v (j_v^d - j_v) \leq k^d - k.$$

All frequencies are nonnegative. Forgetting signs counts these trees by $C = t + C^2/(1 - C)$, which yields (8.3). Since its coefficients are nonnegative and $C(\rho_*) < 1$, one has $c_k \leq \rho_*^{-k}$.

For $\operatorname{Re} u \leq 0$, each exponential has modulus at most one. The geometric majorant proves normal convergence and (8.5). The summed series satisfies $|P_d(u)| \leq C(|u|) < 1$. The tree majorant also bounds all the substitutions in (8.1) by $\sum_{j \geq 2} C(|u|)^j < \infty$. Absolute convergence therefore verifies the identity throughout the stated half-disc. The resulting $Q_d = e^{-u} P_d$ equals the small branch near zero and provides its holomorphic continuation to that half-disc. In particular $|Q_d| \leq e^{\rho_*} C(\rho_*) < 1$, so the original kernel remains absolutely convergent there as well. $\square$

Proposition 8.2 (All-orders boundary-uniform remainder). *Write the formal Taylor series of P_d as $\sum_{n \geq 1} a_{d,n} u^n$. For $\operatorname{Re} u \leq 0$, $r = |u| \leq r_0 < \rho_*$, and $N \geq 2$,*

$$\left| P_d(u) - \sum_{n < N} a_{d,n} u^n \right| \leq r^N M_{d,N}(r_0), \quad (8.6)$$

where

$$M_{d,N}(r_0) = \frac{\rho_*^{-N}}{1 - r_0/\rho_*} + \sum_{k=1}^{N-1} c_k \frac{(k^d - k)^{N-k}}{(N-k)!}. \quad (8.7)$$

For the approximation to Q_d obtained by multiplying the truncated polynomial by e^{-u} , multiply this bound by e^r .

Proof. For $\operatorname{Re} z \leq 0$, the integral Taylor remainder gives

$$\left| e^z - \sum_{m < L} \frac{z^m}{m!} \right| \leq \frac{|z|^L}{L!}.$$

In block $k < N$, take $L = N - k$ and use (8.4). Blocks $k \geq N$ are bounded without expansion. These two sums give (8.7). At any fixed Taylor degree only finitely many blocks contribute, so the truncated coefficients are exactly $a_{d,n}$. $\square$

For $d = 2$ this is the predecessor's quadratic certificate [4], now recovered from the tree formulation. Its validity on the imaginary boundary is essential; estimates only on proper left sectors would not give the same critical-direction conclusion.

9 An explicit optimized scale for every integer degree

Theorem 9.1 (Sharp growth of the positive remainder majorant). *For fixed $d \geq 2$ and $r_0 < \rho_*$,*

$$\log M_{d,N}(r_0) = (d-1)N \log N - dN \log \log N \quad (9.1)$$

$$+ \left[d \log \frac{d}{d-1} - (d-1) \right] N + o(N). \quad (9.2)$$

In particular, for every $H > 0$ there is C_H such that

$$M_{d,N}(r_0) \leq C_H H^N (N!)^{d-1} \quad (N \geq 2). \quad (9.3)$$

Proof. For $2 \leq k < N$, let $T_{N,k} = c_k(k^d - k)^{N-k}/(N-k)!$. The logarithm of a sum of at most N positive terms differs from its maximum by at most $\log N$. Put $t = k/N$. The bounds $1 \leq c_k \leq \rho_*^{-k}$, $k^d - k \leq k^d$, and Stirling's formula, with error uniform in $N - k \geq 1$, give

$$\frac{1}{N} \log T_{N,k} \leq (1-t)\{(d-1) \log N + d \log t - \log(1-t) + 1\} + t \log(\rho_*^{-1}) + O(\log N/N).$$

Set $a = t \log N$. After subtracting $(d-1) \log N - d \log \log N$, the right-hand side becomes

$$d \log a + 1 - (d-1)a + R_d(t) + O(\log N/N),$$

where

$$R_d(t) = -dt \log t - (1-t) \log(1-t) - t + t \log(\rho_*^{-1}).$$

The function R_d is bounded and tends to zero as $t \downarrow 0$. Since $d \log a + 1 - (d-1)a$ tends to minus infinity at both endpoints of $(0, \infty)$, maximizing indices must have a in a fixed compact

subinterval. There $t \rightarrow 0$ uniformly. The limiting maximum is attained at $a = d/(d-1)$ and equals $d \log(d/(d-1)) - (d-1)$.

For a matching lower bound, take $k = \lfloor dN/((d-1) \log N) \rfloor$ and use $c_k \geq 1$. Replacing $k^d - k$ by k^d changes the logarithm by $(N-k) \log(1 - k^{1-d}) = o(N)$, including $d=2$. The same calculation attains the limiting maximum. The geometric term in (8.7) has logarithm $O(N)$ and is smaller. This proves (9.2). Subtracting $(d-1) \log(N!)$ gives $-dN \log \log N + O(N)$, which implies (9.3). $\square$

Corollary 9.2 (Certified optimized truncation). *Set $s = d-1$, $L = \log(1/r)$, and*

$$N_d(r) = \left\lfloor d^{-d/s} r^{-1/s} L^{d/s} \right\rfloor. \quad (9.4)$$

For every $\varepsilon > 0$, all sufficiently small $r > 0$, and uniformly on $|u| = r$, $\operatorname{Re} u \leq 0$,

$$\left| Q_d(u) - e^{-u} \sum_{n < N_d(r)} a_{d,n} u^n \right| \leq \exp \left[- \left(s d^{-d/s} - \varepsilon \right) r^{-1/s} L^{d/s} \right]. \quad (9.5)$$

Proof. Insert (9.4) in (9.2). One has

$$\log N = \frac{L + d \log L - d \log d}{s} + o(1), \quad \log \log N = \log L - \log s + o(1).$$

It follows that $\log(r^N M_{d,N}) = -sN(1 + o(1))$. The factor e^r in Proposition 8.2 is negligible at this scale. The arbitrary ε absorbs the remaining lower-order terms. $\square$

The constant is $1/4$ for $d=2$ and $2/(3\sqrt{3})$ for $d=3$. It is a sharp constant for the leading optimization of this positive majorant. A matching lower bound for the *actual signed truncation error* is not proved. In the transseries coordinate $u = -e^{-X}$, (9.5) is a bound on the scale

$$\exp \left[- \left((d-1) d^{-d/(d-1)} - o(1) \right) e^{X/(d-1)} X^{d/(d-1)} \right].$$

This upper bound alone does not establish a nonzero additional transseries sector on that scale.

10 Fine summability and the angular obstruction

10.1 Conventions and the classical implication used

For $\hat{f}(z) = \sum_{n \geq 0} f_n z^n$, use the integrated Borel transform

$$\mathcal{B}_0 \hat{f}(\zeta) = \sum_{n \geq 0} \frac{f_n}{n!} \zeta^n. \quad (10.1)$$

A series is *finely summable in direction π* if this germ continues to a tube of some fixed width about $(-\infty, 0]$, with a bound $Ce^{H|\zeta|}$ there. The sum on $z = -r$ is

$$\mathcal{S}_\pi \hat{f}(-r) = \frac{1}{r} \int_0^\infty e^{-s/r} \mathcal{B}_0 \hat{f}(-s) ds. \quad (10.2)$$

This is the fine convention of Sauzin's Sections 7 and 9, reflected to the negative ray [9]. By *angular 1-summability near π* we mean continuation to an open Borel sector containing that direction, with exponential bounds uniform on each smaller angular sector. Contour rotation then supplies a sum in a physical sector of opening greater than π .

The Nevanlinna–Sokal theorem says that holomorphy on a tangent disc $\operatorname{Re}(-1/z) > R^{-1}$, together with uniform remainders bounded by $CA^N N!|z|^N$, reconstructs the function by fine Borel summation [8]. We use this classical theorem, not a replacement theorem.

10.2 Recovery of the positive quadratic result

For $d = 2$, (9.3) and Proposition 8.2 imply uniform Gevrey-1 remainders for P_2 on a closed left half-disc. Every sufficiently small tangent disc lies there. Nevanlinna-Sokal therefore identifies P_2 as its fine sum. Multiplication by e^{-u} preserves that class and identifies Q_2 as the fine sum of $\widehat{Q}_2$. Taking a boundary coefficient limit in the all-orders estimate shows that its coefficients satisfy $|[u^n]P_2| \leq C_H H^n n!$ for every $H > 0$. Hence $\mathcal{B}_0 \widehat{P}_2$ is entire. The same holds after multiplication by e^{-u} .

For completeness, the nonlinear closure needed to pass to U_2 can be seen by conjugating a tangent-to-identity series at zero to one at infinity. If $f(z) = z + c + \varphi(z)$, $\varphi = O(z^{-1})$, has a fine-summable Borel minor ϕ , first remove the constant translation. The inverse correction has the formal representation

$$\chi = \sum_{k \geq 1} \frac{(-1)^k}{k!} \partial_z^{k-1}(\varphi^k), \quad \mathcal{B}\chi(\xi) = - \sum_{k \geq 1} \frac{\xi^{k-1}}{k!} \phi^{*k}(\xi). \quad (10.3)$$

If $|\phi(\xi)| \leq M e^{c_0|\xi|}$ on a convex tube, convolution along a straight simplex gives

$$|\phi^{*k}(\xi)| \leq M^k e^{c_0|\xi|} \frac{|\xi|^{k-1}}{(k-1)!}.$$

Thus (10.3) converges normally and is bounded by $M \exp((c_0 + 2\sqrt{M})|\xi|)$. Termwise Laplace integration and the convergent analytic Lagrange formula at sufficiently large $\operatorname{Re} z$ identify its sum as the actual inverse. Translations, reciprocals with nonzero constant term, and multiplication or division by z preserve the fine class, after narrowing the tube for differentiation. Conjugation by $z \mapsto -1/z$ therefore proves the claim for U_2 . These are the half-strip operations also used in the predecessor [4]; the angular version is classical [9].

If the input minors are entire, the same estimates on each fixed compact disc, with constants depending on that disc, prove that the transformed minors are entire. Consequently $\mathcal{B}_0 \widehat{U}_2$ is entire as well. These local compact-disc bounds make no assertion about exponential growth on an unbounded angular sector.

10.3 The negative answer to angular summability

Proof of Theorem 1.2 for the exact quadratic model. The preceding subsection supplies fine summability, entire Borel transforms, and identification of the actual sums. Suppose $\widehat{Q}_2$ were angularly 1-summable near π . Its angular sum agrees with its fine sum along the negative ray, because both use the same continued Borel germ in (10.2). Contour rotation makes the angular sum holomorphic in a sector about the negative axis of opening greater than π . The identity theorem identifies it with Q_2 on their left-hand overlap. It would thus continue Q_2 through small points of its imaginary natural boundary, contradicting Theorem 1.1.

If $\widehat{U}_2$ were angularly summable, the classical closure of angularly summable tangent-to-identity series under inversion would make $\widehat{Q}_2$ angularly summable, which has just been excluded. Alternatively, its fine sum is already U_2 , and the same sector argument meets the natural boundary Γ_2 directly. $\square$

Corollary 10.1 (Entire Borel transforms with no angular exponential bound). *For either $B = \mathcal{B}_0 \widehat{Q}_2$ or $B = \mathcal{B}_0 \widehat{U}_2$, for every $\varepsilon > 0$ and every $C, H > 0$, there are arbitrarily large ζ with $|\arg \zeta - \pi| < \varepsilon$ (using the argument centered at π) such that*

$$|B(\zeta)| > C e^{H|\zeta|}.$$

Thus the obstruction is growth at Borel infinity, not a finite singularity of these entire Borel transforms.

Proof. A uniform exponential bound on such a sector, together with entireness, would give angular summability. If the bound held beyond some radius, boundedness on the remaining compact disc would give a global bound after increasing C . That too would give angular summability. Thus violations occur at arbitrarily large points. $\square$

The conclusion does not classify growth on each individual nonreal ray. Failure of a sector-wide bound is stronger than a failure at one fixed finite point, but weaker than a pointwise growth law on every ray.

10.4 Completion for finitely modified quadratic slopes

Normalize the small inverse by $P = e^{\lambda_1 u} Q$. Its tree frequency at an internal vertex of arity j is $\lambda_j - j\lambda_1$. For slopes equal to j^2 eventually, there is a constant C such that a k -leaf tree has frequency β satisfying

$$-Ck \leq \beta \leq Ck^2.$$

Indeed $\sum(j-1) = k-1$, the number of internal vertices is at most $k-1$, and only finitely many arities differ from the quadratic rule. On $|u| \leq r_0$, $\operatorname{Re} u \leq 0$, negative frequencies cost at most e^{Ckr_0} . This merely changes the geometric tree-count constant. The proof of Proposition 8.2 now uses $(Ck^2)^{N-k}$ and an adjusted geometric constant. The saddle estimate still gives

$$\log M_N = N \log N - 2N \log \log N + O(N),$$

so the remainders are Gevrey-1 of zero type. The same summation and inverse arguments apply. Corollary 6.3 supplies the natural boundary, hence excludes angular summability. This proves the remaining assertion of Theorem 1.2.

11 Further extensions and sharp limitations

11.1 Affine-quadratic exponents

The natural-boundary mechanism also treats

$$\lambda_j = Aj^2 + Bj + C, \quad A > 0,$$

with real B, C , near the origin. Set $w = Au$ and $z = qe^{Bu}$. Then

$$\sum_{j \geq 1} q^j e^{\lambda_j u} = e^{Cu} \Theta_2(z, w).$$

The level equation becomes $\Theta_2(z, w) = (w/A)e^{-Cw/A}$, an analytic nonzero forcing to which Theorem 6.1 applies. Multiplication of the moving root by a nonvanishing holomorphic factor and the positive linear change of u cannot remove the boundary. Finite perturbations are again permitted.

11.2 The essential small-parameter hypothesis

The local theorem assumes $q_0 \neq 0$. At $q_0 = 0$ the entire kernel vanishes, and the holomorphic curve $q \equiv 0$ is an obvious exception. The natural boundary proof does not overlook it: it proves nonextension on a dense set of nonzero-root points and then uses openness of continuation to reach the isolated zero points, including the origin.

For $d \geq 3$, the smallness hypothesis separates the d th powers of nearest poles. Without it, cancellation of a leading nearest-pole pair is a real possibility in the asymptotic formula. One must then inspect further orders or further poles. The present theorem does not silently assume that such a pair cannot occur.

11.3 No contradiction with local resurgence of other theta expansions

The partial-theta literature studies several different expansions, including ones at rational boundary points with periodic amplitudes, and their Borel singularities [10]. Our coefficient asymptotics concern a geometric weight q_0^j and a holomorphically moving q , and the nonlinear conclusion concerns the inverse near the vanishing level. Neither the expansion point nor the summation problem may be exchanged without proof.

In particular, an entire Borel transform is endlessly continuable in the finite plane in the simplest sense, while still failing the growth requirements for angular Laplace summation. The word “resurgent” alone therefore does not answer the angular question settled here.

12 Verification, reproducibility, and proof audit

12.1 Exact finite checks

The supplied `code/verify.py` uses rational arithmetic for three independent constructions: the original feedback recurrence, formal compositional inversion, and the exponential-block tree recurrence. At the recorded default order 24, for $d = 2, 3, 4$, it performed **324 exact comparisons**. Both inverse compositions, the inverse kernel identity, the displayed low coefficients, block support, block mass, and block-to-Taylor conversion passed.

The block calculations use blocks through degree 10. The ordinary formal series calculations use degree 24. These are deliberately separate checks, not a single recurrence compared with itself. The CSV files retain exact rational coefficients. None of these finite checks proves a natural boundary or an infinite-order asymptotic theorem.

12.2 High-precision diagnostics

The optional diagnostic section ran at 110 decimal digits with `mpmath`. It checked the moving-curve asymptotics at ten (d, n) pairs, four rational boundary roots, 352 quadratic Fourier magnitudes, and a tied-pole mean-square example. These are floating-point diagnostics, not interval certificates.

For $q(v) = q_0 \exp(3v/7 + v^2/9)$ with constant periodic coefficient $c_j = 1$, the ratio of the exact finite coefficient formula to its predicted leading term was as follows. The quadratic case uses $q_0 = 1/5$; the cubic case uses $q_0 = 1/30$.

n	$d = 2$ ratio	$d = 3$ ratio
8	1.003554662	1.075228530
16	1.001774139	1.035862320
32	1.000884142	1.017556494
64	1.000441140	1.008690883
80	1.000352752	1.006939094

These values are consistent with the limits; the proof of the limits is Section 4, not this table.

At $u = 2\pi i/b$, the boundary equation is finite and rational in q :

$$\Theta_2(q, 2\pi i/b) = \frac{\sum_{j=1}^b e^{2\pi i j^2/b} q^j}{1 - q^b}. \quad (12.1)$$

Solving it for the small root gives the following diagnostic for the quadratic displacement in (7.1):

b	$t = 2\pi/b$	$\operatorname{Re} Q_2(it)/t^2$
127	0.04947390006	1.998438396
257	0.02444819186	1.999605886
509	0.01234417546	1.999898701
1021	0.00615395231	1.999974770

The maximum high-precision residual was below $1.4 \cdot 10^{-114}$. This is a residual in the stated floating-point computation, not a certified error bound of that magnitude for every printed digit.

12.3 The proof's load-bearing points

The natural-boundary theorem rests on four analytic facts proved above: the pole expansion of the moment generating function; a global factorial majorant valid even near the end of the coefficient sum; a noncancellation argument when two nearest poles tie; and a contradiction obtained only *after assuming* that the implicit root is holomorphic across a point. A fixed-parameter gap theorem, numerical derivative growth, or a plot of boundary values would not replace any of these steps.

The positive quadratic summation result additionally uses the classical Nevanlinna–Sokal theorem and the standard convolution/Lagrange inverse mechanism. The new natural-boundary proof itself does not depend on a summability theorem, the accuracy of the numerical tables, or an unverified claim about the larger transseries field.

12.4 Suggested formalization layers

A realistic formalization would separate finite Fourier inversion and finite coefficient identities from complex analytic estimates. First formalize (4.4), (4.7), and (4.9). Next package the Cauchy principal-part estimate and dominated-convergence argument. Only then assemble the periodic boundary obstruction and its implicit-function corollary. The summability corollary is a later, independent interface. No Lean source is included or represented as checked in this package.

13 Further research questions and topics

The following problems are not used as premises in the proofs.

13.1 Boundary jets of the actual implicit solution

Question 13.1. At a fixed nonzero rational imaginary point u_0 , determine the precise large-order asymptotics of the actual boundary derivatives $Q_d^{(n)}(u_0)/n!$, including constants and cancellations.

The moving-curve theorem assumes a holomorphic curve and proves that such a curve cannot be the implicit root. It does not give the root's own large-order law. A nonlinear asymptotic inversion of the periodic moment system is the next step. Establishing exact boundary Gevrey order would quantify how far the C^∞ boundary is from analyticity.

13.2 The angular growth of the entire Borel transform

Question 13.2. For $B = \mathcal{B}_0\widehat{Q}_2$ or $\mathcal{B}_0\widehat{U}_2$, determine the growth as $|\zeta| \rightarrow \infty$ on rays adjacent to the negative axis, and identify the largest curvilinear regions carrying exponential bounds.

Corollary 10.1 rules out every fixed angular neighborhood, but leaves room for widening tubes and parabolic regions. An explicit transition law could explain how fine summation approaches its angular failure and improve numerical contour choices.

Editorial note (ProveIt, 2026-09-29). For $B = \mathcal{B}_0\hat{Q}_2$ the maximum modulus on circles is known: `Signed_Quadratic_Feedback_Inversion/` [6], Theorem `thm:borel`, gives $\max_{|\zeta|=R} |B(\zeta)| \sim -B(R)$, with an explicit equivalent, which bounds the growth on every ray. The growth on individual rays adjacent to the negative axis is not determined there.

13.3 Higher-degree pole collisions

Question 13.3. Remove the convenient bound $|q_0| < e^{-d}$ from Proposition 5.2 for $d \geq 3$, or classify the exceptional curves on which leading d th-power pole collisions cancel.

The exact obstruction is equality of nearest-pole d th powers, not a failure of the coefficient identity. Subleading terms and additional pole circles should be analyzed before asserting a general nonextension theorem for the whole q -disc.

13.4 General polynomial and nonpolynomial slopes

Question 13.4. Extend the natural-boundary theorem from j^d and affine quadratics to arbitrary positive-leading-coefficient integer polynomials, and then to nonintegral powers or asymptotically polynomial slopes.

For integer polynomials, rational boundary phases remain periodic, but the lower polynomial terms alter the moment asymptotics. For noninteger powers, periodicity is lost; a different replacement for the finite Fourier pole spectrum is needed. Merely knowing coefficient Gevrey growth does not settle analytic continuation.

13.5 Actual optimal errors rather than positive majorants

Question 13.5. Does the signed error for optimally truncated Q_d attain the scale and constant in (9.5) along some nonexceptional ray? If not, what is its true leading scale?

A positive tree majorant discards cancellations. Matching lower bounds would require a direct analysis of the signed blocks, not another optimization of the same upper estimate.

Editorial note (ProveIt, 2026-09-29). For $d = 2$ the least formal term of $\hat{Q}_2$ on the positive axis is determined by `Signed_Condensation_Sharp_Quadratic_Reversion/` [7], Theorem `thm:least`: at $a = 1$ it is $\exp(-(1/4 + o(1)) \log^2(1/t)/t)$, with the constant $1/4$ of (9.5) at $d = 2$. That is a statement about formal terms; it gives no lower bound for the analytic truncation error, so the question stays open.

13.6 Global boundary geometry and inverse branches

Question 13.6. Continue the canonical root along the left half-plane as far as possible. Locate any interior critical points or poles and determine the global image of the imaginary boundary under Q_d .

The current theorem is local near zero. A global branch may encounter critical points before the geometric-series condition $|Q_d| < 1$ fails. The local natural boundary is proved; its global continuation and topology are separate questions.

13.7 Summation procedures for higher-degree feedback

Question 13.7. For $d \geq 3$, identify a summation procedure that reconstructs the canonical literal branch from its Gevrey- $(d-1)$, logarithmically improved series, without assuming an angular continuation which the branch cannot have.

The closed-half-disc remainder and convergent block expansion already give an evaluation procedure. A generalized fine summation or acceleration theorem would provide a transform-side characterization. It must distinguish an arbitrary function with the same formal expansion from the branch selected by the literal infinite feedback equation.

13.8 Effective boundary certificates

Question 13.8. Turn the moment and mean-square proof into a finite, certified procedure that rules out a proposed analytic continuation radius or derivative bound.

At rational boundary points the pole set is explicit. Quantitative errors in Theorem 4.1, together with an effective lower bound for a short mean-square average, would convert the qualitative contradiction into machine-checkable analytic certificates.

14 Conclusion

The infinite feedback equation supplies a concrete separation between formal inversion, smooth boundary realization, fine Borel reconstruction, and angular analytic continuation. In the quadratic model, fine reconstruction works and the Borel transforms are entire, yet every angular enlargement of the negative summation direction fails. The obstruction survives finite-core changes and analytic forcing because a moving holomorphic parameter cannot remove the factorial moment contribution of the periodic boundary spectrum.

The same proof gives natural boundaries for all integer-power feedback kernels j^d , while the tree expansion gives usable, uniform remainder bounds on the permitted side. The result is therefore not simply an obstruction: it identifies both what continuation cannot do and what certified asymptotic evaluation can still accomplish.

A Reproduction commands and package contents

The article source is standalone and uses an embedded bibliography. From the package directory, run

```
sh build.sh
python code/verify.py --order 24 --exact-only
python -m pip install -r requirements.txt
python code/verify.py --order 24
```

The first verification command needs only the Python standard library. The second full run additionally uses mpmath. Running the program replaces its own data files; an exact-only run marks diagnostics as not run rather than pretending they passed. No program accesses the network or changes the upstream repository.

The package includes source and PDF, the verification program, exact coefficient and block files, numerical diagnostic tables, a recorded run, source-provenance notes, and a proof-status audit. PDF compilation and rendering checks are recorded separately from mathematical verification.

References

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